

Application
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APPLICATION:

1. The position, in metres, m , at time t , in seconds, of an object moving along a horizontal line is given by $s(t) = 2t^4 - 16t^3 + 20t^2 - 7$ for $0 \leq t \leq 6$.
- When is the object at rest?
 - What is the velocity of object at 2 seconds?
 - Is the object speeding up or slowing down at 2 seconds?

a) $v(t) = 8t^3 - 48t^2 + 40t$ Object at rest at $0s, 1s, 5s$
 $= 8t(t^2 - 6t + 5)$
 $0 = 8t(t-1)(t-5)$
 $t = 0, 1, 5$

b) $v(2) = 2(16) - 16(8) + 20(4) - 7$
 $= 32 - 128 + 80 - 7 = -23 \text{ m/s}$
 $\therefore$ velocity at 2s is -23 m/s

c) Find by finding $a(2)$
 $a(t) = 24t^2 - 96t + 40$
 $a(2) = 24(4) - 96(2) + 40$
 $= 96 - 192 + 40 = -56 \text{ m/s}^2$

$\therefore$ A negative velocity and acceleration at 2s shows it is speeding up in the negative direction.

2. Find the equation of the line that is normal to the curve $x^2y = y^2 - 6x$ at the point $(2, 6)$.

$2xy + x^2 \frac{dy}{dx} = 2y \frac{dy}{dx} - 6$ At $(2, 6)$
 $(2x^2 \frac{dy}{dx} - 2y \frac{dy}{dx} = -2xy - 6$
 $\frac{dy}{dx} = \frac{-2xy - 6}{x^2 - 2y}$
 $y = \frac{4}{15}x + b$
 $b = -\frac{8}{15} + b$
 $y = \frac{4}{15}x + \frac{98}{15}$

3. The concentration of a drug in a patient's bloodstream is $C(t) = \frac{5t}{2t^2 + 7}$, where C is in

milligrams per cubic centimetre and t is the number of hours after the drug is taken. Find

the maximum and minimum concentrations for the interval $1 \leq t \leq 4$.

$C'(t) = \frac{5(2t^2 + 7) - 5t(4t)}{(2t^2 + 7)^2} = \frac{10t^2 + 35 - 20t^2}{(2t^2 + 7)^2} = \frac{-10t^2 + 35}{(2t^2 + 7)^2}$

$0 = -10t^2 + 35$
 $10t^2 = 35$
 $t^2 = \frac{7}{2}$
 $t = \sqrt{\frac{7}{2}} \text{ h}$
 (no negative)
 domain

$C(1) = \frac{5}{2+7} = \frac{5}{9} = 0.56 \text{ mg/cm}^3$
 $C(\sqrt{\frac{7}{2}}) = \frac{9.35}{14} = 0.67 \text{ mg/cm}^3$
 $C(4) = \frac{20}{32+7} = \frac{20}{39} = 0.51 \text{ mg/cm}^3$

$\therefore$ The max concentration is about 0.67 mg/cm^3 at $\sqrt{\frac{7}{2}}$ or 1.87 seconds. The min is about 0.51 mg/cm^3 at 4 hours.

1. The position function of an object moving horizontally along a straight line as a function of time is $s(t) = 8t^3 - 4t^2$, in metres, at time, t , in seconds. Determine the acceleration of the object at $t = 4$ s.

- a. 184 m/s²
b. 248 m/s²

- c. 56 m/s²
d. 352 m/s²

$$v(t) = 24t^2 - 8t$$

$$a(t) = 48t - 8$$

$$a(4) = 192 - 8 = 184 \text{ m/s}^2$$

2. If the equation $s(t) = t^3 - 2t^2 + 1$ describes an object's motion in metres, at what time t , in seconds, will the object be stationary?

- a. 1 s
b. 0 s and $\frac{4}{3}$ s

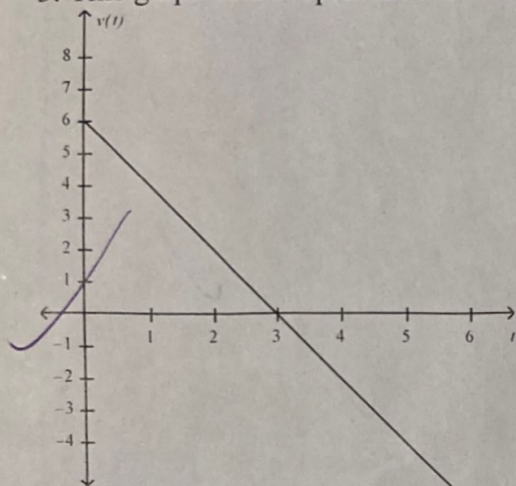
- c. 0 s
d. $\frac{4}{3}$ s

$$v(t) = 3t^2 - 4t$$

$$0 = t(3t - 4)$$

$$t = 0, \frac{4}{3}$$

3. This graph could represent the velocity of which of the following position functions?



a) $v(t) = -2t + 6 \rightarrow$

b) $v(t) = 2t + 6 \rightarrow$

c) $v(t) = -2 \rightarrow$

d) $v(t) = 4t + 6 \rightarrow$

$\therefore$ a)

- a. $s(t) = -t^2 + 6t + 7$
b. $s(t) = t^2 + 6t + 1$

- c. $s(t) = -2t + 6$
d. $s(t) = 2t^2 + 6t$

4. Let the position of a certain particle be described by the function $s(t) = kt^2 - (5k + 1)t + k$. For which constant value of k is the particle stationary when the time $t = 2$ s?

- a. $k = -\frac{1}{5}$
b. $k = 1$

- c. $k = 2$
d. $k = -1$

$$s(t) = kt^2 - 5kt - t + k$$

$$v(t) = 2kt - 5k - 1$$

$$v(2) = 0$$

$$0 = 2k(2) - 5k - 1$$

$$0 = 4k - 5k - 1$$

$$1 = -k$$

$$k = -1$$

check

$$v(t) = -2t + 5 - 1 = -2t + 4$$

$$v(2) = -4 + 4 = 0 \checkmark$$

$\therefore$ d)